

22-S2-Q3

(a) method + procedure

(b) 400 $\begin{matrix} \swarrow & 1 \\ \searrow & -1 \end{matrix}$

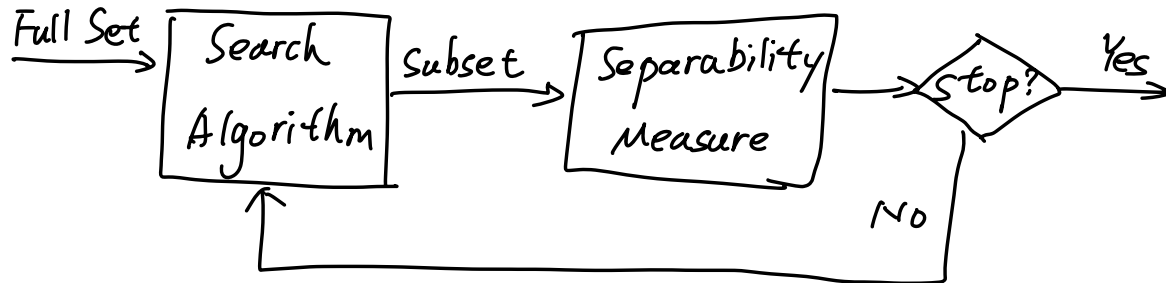
m_1 m_2 s_1 s_2

Fisher linear discriminant

(c) repeated k-fold

Solution ① The filter method

② procedure map



③ Search Algorithm can be

- (1) Exhaustive search method
- (2) Sequential forward selection
- (3) Sequential backward elimination

④ Separability measures can be

- (1) Mahalanobis distance-based separability measure
- (2) Scatter-based separability measure

⑤ the stopping criterion could be

- (1) number of features to be selected is reached
- (2) separability measure no longer improves
- (3) Others

⑥ Example

Sequential forward feature selection algorithm

(1) In step 1, consider each feature as a candidate

$$F^{(i)} = \{x_i\} \quad i=1, 2, \dots, n$$

evaluate the separability measure

$$J^{F(i)} = \frac{\text{Tr}(S_B^{F(i)})}{\text{Tr}(S_W^{F(i)})}$$

The feature that has the largest separability is selected as the first feature

$$k = \arg \max (J^{F(i)})$$

$$z_1 = x_k$$

(2) In step 2, consider each of the

remaining $n-1$ features as the candidate for the second feature

$$F^{(i)} = \{z_1, x_i\}, \quad i \neq k$$

Evaluate the separability measure

$$J^{F(i)} = \frac{\text{Tr}(S_B^{F(i)})}{\text{Tr}(S_W^{F(i)})}$$

The feature that has the largest separability is selected as the second feature

(3) The selection process is continued until the stopping criterion is satisfied

$$(b) \textcircled{1} S_w = S_1 + S_2$$

$$W^* = S_w^{-1} (m_1 - m_2)$$

$$b^* = - \frac{(W^*)^T (m_1 + m_2)}{2}$$

$$g(x) = W^T x + W_0$$

$$= (W^*)^T x + b^*$$

If $g(x) > 0$, classifying x to class 1

If $g(x) < 0$, classifying x to class -1

If $g(x) = 0$, x on the decision boundary

$$\textcircled{2} W^* = S_w^{-1} (m_1 - m_2)$$

$$= \begin{bmatrix} -0.005344 \\ -0.01077 \\ -0.01089 \end{bmatrix}$$

$$b^* = - \frac{(W^*)^T (m_1 + m_2)}{2}$$

$$= 0.001298$$

③ we denote $\pi = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$$g(x) = -0.005344 x_1 - 0.01077 x_2 - 0.01089 x_3 \\ + 0.001298$$

cc) ① k -fold

In k -fold cross validation, the data set is divided into k -completely distinct or non-overlapping random partitions called folds.

Each of the k folds is given an opportunity to be used as a held back test set, whilst all other folds collectively are used as a training dataset.

A total of k models are fit and evaluated on the k hold-out test sets and the mean performance is reported.

② Repeated k -fold

The estimate of model performance via k -fold cross validation can be noisy.

One solution to the problem is repeat the k -fold multiple times and report the mean performance across all folds and

all repeats,

③ procedure

(1) step repeat = 1

(2) k fold & evaluation

(3) repeat = repeat + 1
reshuffle data

(4) Go to step (2) $\rightarrow$ total number = 50

(5) average performance